Business Insight Report

Student Performance Dataset Analysis

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1. Objective

The objective of this project is to analyze student performance data, identify patterns and trends, and generate actionable insights through data cleaning, visualization, and exploratory analysis.

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2. Dataset Summary

The dataset contains student demographic information, attendance records, study habits, and academic performance indicators.

- Original Records: 206
- Cleaned Records: 189
- Numerical Features: Age, Attendance Percentage, Study Hours Per Week, Assignment Score, Midterm Score, Final Score
- Categorical Features: Gender, Parental Education, Internet Access, Extra Activities

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3. Analysis Approach

The analysis was conducted using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn.

Steps performed:

- Data inspection
- Missing value handling
- Duplicate removal
- Invalid value correction
- Descriptive statistics
- Data visualization
- Correlation analysis

4. Key Findings

1. Most students scored between 40 and 80 marks in the final examination.
2. The average final score was approximately 59 marks.
3. Male and female students demonstrated similar academic performance.
4. Study hours showed only a weak relationship with final scores.
5. Correlation analysis indicated that no single variable strongly predicts academic performance.
6. Student performance appears to be influenced by multiple factors.

5. Recommendations

1. Provide additional support to low-performing students.
2. Encourage consistent study habits and academic engagement.
3. Improve data collection processes to reduce missing and inconsistent records.
4. Monitor multiple performance indicators rather than relying on a single metric.
5. Conduct further analysis using larger datasets.

6. Limitations

- Limited dataset size after cleaning.
- Some potentially important variables were unavailable.
- Correlation does not imply causation.
- Findings are specific to the available dataset.

7. Conclusion

The project successfully demonstrated data cleaning, exploratory analysis, visualization, and business insight generation. The results indicate that student performance is influenced by multiple factors, emphasizing the importance of data-driven decision-making and proper data preparation before analysis.